

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013–14

Paper: CSB-601 — Database Management System

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions including Question 1 which is compulsory. Terms and symbols used have their standard meaning.*

Question 1. Fill in the blanks: [10]

- (a) The number of attributes in a relation is called the _____ of the relation.
- (b) Immunity of the external schemas to changes in the internal schema is referred to as _____.
- (c) When one column of a table refers to the values in another column of the same table, it introduces a _____ integrity constraint.
- (d) The virtual page table never changes during the execution of a transaction — this property is called _____.
- (e) Along with the **GROUP BY** clause, one can use the _____ clause to define a condition on the group of tuples.
- (f) _____ and _____ are two unary operations in relational algebra.
- (g) 4NF is based on the concept of _____ dependency.
- (h) A schedule S is not conflict serializable if the precedence graph contains _____.
- (i) _____ and _____ are the two desirable properties of relational schema decomposition.
- (j) _____ is an abstraction through which relationships are treated as higher-level entities.

Question 2. [3 + 4 + 3 + 5 = 15]

- (a) Explain the difference between primary key and superkey.
- (b) What are multi-valued attributes, and how can they be handled within the database design?
- (c) What is meant by a recursive relationship type? Give an example.
- (d) Define cardinality ratio. Draw an ER diagram to illustrate different cases of cardinality ratio.

Question 3. [6 + 6 + 3 = 15]

- (a) What is the lossless-join property of decomposition? Suppose that we decompose the schema $R = (A, B, C, D, E)$ into $R_1 = (A, B, C)$ and $R_2 = (A, D, E)$. Check whether this decomposition is lossy or lossless if the following set F of functional dependencies holds: $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$.
- (b) Define full, partial, and transitive functional dependencies. Explain the normal form based on transitive functional dependency with a suitable example.
- (c) Define multi-valued functional dependency.

Question 4.

[6 + 6 + 3 = 15]

- (a) Discuss the immediate update technique of recovery. How is it different from the deferred update technique? Why is it called the UNDO/REDO method?
- (b) What is the two-phase locking protocol? What are some variations of the two-phase locking protocol? Why is strict two-phase locking often preferred?
- (c) What is a recoverable schedule? Why is recoverability of schedules desirable?

Question 5.

[5 + 10 = 15]

- (a) With a suitable example, explain the difference between natural join, equi-join, and Cartesian product operations.
- (b) Consider the following relational database. Answer each of the queries in SQL:

Branch(bname, bcity, assets)
Customer(cname, street, ccity)
Depositor(cname, account#)
Account(bname, account#, balance)

- (i) Find account numbers of accounts with balances between Rs. 70,000 and Rs. 1,00,000.
- (ii) Find all customers whose street includes the substring "main".
- (iii) Find branches and their average balances where the average balance is more than Rs. 2,000.
- (iv) Delete the records of all accounts with balances below the average.
- (v) Increase all balances by 10% whose balance is greater than the average.

Question 6.

[6 + 3 + 6 + 5 + 4 + 6 = 30]

- (a) Describe the ANSI/SPARC three-level architecture. How are these three levels related to the concept of physical and logical data independence?
- (b) Describe the steps of query processing with a flow diagram.
- (c) Explain any three heuristic rules for query optimization through examples.
- (d) Define weak entity type. Explain how it is handled in relational database designing.
- (e) Explain the Select and Project operators in Relational Algebra through an example.

- (f) What are the database security issues? What is the difference between discretionary and mandatory access control?